Revision Notes

CBSE Class 12 Mathematics • NCERT Part I, Chapters 1–6

Relations and Functions • Inverse Trigonometric Functions • Matrices
Determinants • Continuity and Differentiability • Application of Derivatives

How to use these notes. Keep the separate **formula sheet** next to you. Each chapter has four parts:

- **A JEE basics (extra)**: purple boxes; not needed for the board exam.
- **B Exam tips**: the "which method?" checklist, mistakes that lose marks, and question types.
- **C Exercise-wise solution patterns**: every NCERT exercise grouped into a few patterns, each with one solved question from the book. Try the solved question yourself first, then check. The small line under each exercise lists which questions belong to which pattern.
- **D Pattern spotter**: a one-look table for final revision. Cover the middle column and say the method aloud.

Question numbers match the NCERT textbook (Reprint 2026–27).

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Chapter 1 Relations and Functions

1A JEE basics (extra)

J1. Counting relations and functions

For $|A| = m$, $|B| = n$: relations from A to B: 2^{mn}; functions: n^m; one-one functions ($m \le n$): $\dfrac{n!}{(n-m)!}$; bijections ($m = n$): $n!$.

Reflexive relations on a set of n elements: 2^{n^2-n}; symmetric: $2^{n(n+1)/2}$.

Example: onto functions from $\{1, \ldots, n\}$ to itself $= n!$ (Misc Q4), because on a finite set onto $\iff$ one-one.

J2. Composition and inverse (not in the current NCERT book)

$(g \circ f)(x) = g(f(x))$. f is invertible $\iff$ f is a bijection; to find f^{-1}, solve $y = f(x)$ for x.

Example: $f(x) = \dfrac{x-2}{x-3}$: $y(x-3) = x - 2 \Rightarrow x = \dfrac{3y-2}{y-1}$, so $f^{-1}(y) = \dfrac{3y-2}{y-1}$, $y \neq 1$.

J3. Quick tests

- A strictly increasing or strictly decreasing function is one-one. For a graph: horizontal line cuts it at most once.
- $f : \mathbf{R} \to \mathbf{R}$ is onto $\iff$ range $= \mathbf{R}$. Even functions (x^2, $|x|$, $\cos x$) are never one-one on $\mathbf{R}$.
- Polynomials of odd degree are onto $\mathbf{R}$; of even degree are never onto $\mathbf{R}$.

1B Exam tips

Which method? (check in this order)

1. **Relation questions:** test reflexive, symmetric, transitive *separately*. To prove a property, give a general argument; to disprove, give *one* counter-example.
2. **Equivalence class $[a]$:** list all elements related to a.
3. **One-one:** assume $f(x_1) = f(x_2)$ and show $x_1 = x_2$ (or give two inputs with the same output to disprove).
4. **Onto:** take any y in the co-domain, solve $y = f(x)$, and check that x lies in the domain.
5. **Finite sets:** for $f : X \to X$, one-one $\iff$ onto.

Common mistakes that lose marks

- Checking transitivity on only one triple; you need all pairs $(a, b), (b, c)$ in R.
- Ignoring the domain/co-domain: x^2 is one-one on $\mathbf{N}$ but not on $\mathbf{Z}$ or $\mathbf{R}$.
- For onto, the x you find must belong to the domain (e.g. not a fraction when the domain is $\mathbf{N}$).
- "Not symmetric" needs a specific pair $(a, b) \in R$ with $(b, a) \notin R$.

Question types: 1-mark MCQs on the type of relation or function; 3–5 marks: prove an equivalence relation and find classes, prove a function is bijective.

1C Exercise-wise solution patterns

Exercise 1.1 (16 questions, 4 patterns)

Pattern A: Check reflexive / symmetric / transitive

Solved (Q4): R $= \{(a,b) : a \le b\}$ on $\mathbf{R}$. Reflexive: $a \le a$ ✓. Transitive: $a \le b$, $b \le c \Rightarrow a \le c$ ✓. Not symmetric: $(1,2) \in R$ but $2 \not\le 1$, so $(2,1) \notin R$.

Pattern B: Prove an equivalence relation and find classes

Solved (Q8): R $= \{(a,b) : |a-b|$ is even$\}$ on $\{1,2,3,4,5\}$.
Reflexive: $|a-a| = 0$ even. Symmetric: $|a-b| = |b-a|$. Transitive: $a-b$ and $b-c$ even $\Rightarrow a-c = (a-b)+(b-c)$ even.
Classes: $\{1,3,5\}$ and $\{2,4\}$; no element of one is related to the other (odd − even is odd).

Pattern C: Construct examples with given properties

Solved (Q10(i)): symmetric but neither reflexive nor transitive: on $\{1,2,3\}$, R $= \{(1,2),(2,1)\}$. Not reflexive ($(1,1) \notin R$); not transitive ($(1,2),(2,1) \in R$ but $(1,1) \notin R$).

Pattern D: MCQs on a listed relation

Solved (Q16): R $= \{(a,b) : a = b-2,\ b > 6\}$: $(6,8)$ satisfies $6 = 8-2$, $8 > 6$ **(C)**.

Pattern A: Q1–6. Pattern B: Q7, 8, 9, 11–14. Pattern C: Q10. Pattern D: Q15, 16.

Exercise 1.2 (12 questions, 3 patterns)

Pattern A: One-one / onto for formula functions

Solved (Q7): (i) $f(x) = 3-4x$: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (one-one); for any y, $x = \frac{3-y}{4} \in \mathbf{R}$ (onto): **bijective**.
(ii) $f(x) = 1+x^2$: $f(1) = f(-1)$, not one-one; 0 has no pre-image ($1+x^2 \ge 1$), not onto.

Pattern B: Special functions ($[x]$, $|x|$, signum)

Solved (Q3): $f(x) = [x]$: $f(1.2) = f(1.9) = 1$, so not one-one; 0.5 is not an integer, so it has no pre-image: not onto.

Pattern C: Prove bijective (piecewise, products of sets, rational)

Solved (Q10): $f(x) = \dfrac{x-2}{x-3}$ on $\mathbf{R} - \{3\} \to \mathbf{R} - \{1\}$.
One-one: $\dfrac{x_1-2}{x_1-3} = \dfrac{x_2-2}{x_2-3} \Rightarrow x_1 = x_2$ (cross-multiply). Onto: $y = \dfrac{x-2}{x-3} \Rightarrow x = \dfrac{3y-2}{y-1}$, defined for $y \ne 1$ and never equal to 3. Bijective.

Pattern A: Q1, 2, 6, 7, 11, 12. Pattern B: Q3, 4, 5. Pattern C: Q8, 9, 10.

Miscellaneous Exercise (7 questions, 3 patterns)

Pattern A: Bijection proofs

Solved (Q1): $f(x) = \dfrac{x}{1+|x|}$. For $x \geq 0$, $f(x) = \frac{x}{1+x} \in [0,1)$; for $x < 0$, $f(x) = \frac{x}{1-x} \in (-1,0)$. Same sign on each side, and each piece is strictly increasing: one-one. For $y \in [0,1)$, $x = \frac{y}{1-y}$; for $y \in (-1,0)$, $x = \frac{y}{1+y}$: onto.

Pattern B: Relations on sets, equality of functions

Solved (Q3): $A\,R\,B \iff A \subset B$ on $P(X)$: reflexive and transitive, but not symmetric ($\phi \subset X$, $X \not\subset \phi$): not an equivalence relation.

Pattern C: Counting

Solved (Q7): equivalence relations on $\{1,2,3\}$ containing $(1,2)$: classes $\{1,2\},\{3\}$ or $\{1,2,3\}$: 2 **(B)**.

Pattern A: Q1, 2. Pattern B: Q3, 5. Pattern C: Q4, 6, 7.

1D Pattern spotter

If you see...	Do this	See
Relations		
Is R reflexive / symmetric / transitive?	Test each; counter-example to disprove	1.1 A
Prove equivalence, find $[a]$	Three proofs; list related elements	1.1 B
"Give an example of a relation which is..."	Small set, add only needed pairs	1.1 C
Functions		
One-one?	$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$	1.2 A
Onto?	Solve $y = f(x)$; x must be in the domain	1.2 A
$[x]$, $\lvert x\rvert$, sgn	Many-one and not onto on $\mathbf{R}$	1.2 B
Rational or piecewise function	Cross-multiply; solve each piece	1.2 C
Finite set to itself	One-one $\iff$ onto; $n!$ bijections	Misc C

Chapter 2 Inverse Trigonometric Functions

2A JEE basics (extra)

J1. Sum and difference formulas

$\tan^{-1}x+\tan^{-1}y=\tan^{-1}\dfrac{x+y}{1-xy}$ $(xy<1)$; $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\dfrac{x-y}{1+xy}$ $(xy>-1)$.

$\sin^{-1}x+\cos^{-1}x=\tan^{-1}x+\cot^{-1}x=\sec^{-1}x+\operatorname{cosec}^{-1}x=\dfrac{\pi}{2}$.

Example: $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}=\tan^{-1}\dfrac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{6}}=\tan^{-1}1=\dfrac{\pi}{4}$.

J2. Double-angle forms

$2\tan^{-1}x=\sin^{-1}\dfrac{2x}{1+x^2}=\cos^{-1}\dfrac{1-x^2}{1+x^2}=\tan^{-1}\dfrac{2x}{1-x^2}$ (for $|x|<1$); $3\sin^{-1}x=\sin^{-1}(3x-4x^3)$.

J3. Values outside the principal range

$\sin^{-1}(\sin\theta)=\theta$ only for $\theta\in[-\frac{\pi}{2},\frac{\pi}{2}]$. Otherwise rewrite: $\sin\frac{3\pi}{5}=\sin\frac{2\pi}{5}$, so $\sin^{-1}(\sin\frac{3\pi}{5})=\frac{2\pi}{5}$.
$\cos^{-1}(\cos\theta)$: bring θ into $[0,\pi]$; $\tan^{-1}(\tan\theta)$: into $(-\frac{\pi}{2},\frac{\pi}{2})$.

2B Exam tips

Which method?

1. **Principal value:** find the angle with that value *inside* the principal range.
2. $f^{-1}(f(\theta))$**:** check if θ is in the principal range; if not, use periodicity/symmetry to move it in.
3. **Simplify an expression:** substitute $x=\tan\theta$ (for $1+x^2$), $x=\sin\theta$ ($\sqrt{1-x^2}$), $x=a\sin\theta$ ($\sqrt{a^2-x^2}$), $x=\cos\theta$ ($1\pm x$).
4. $\tan(\sin^{-1}a+\cot^{-1}b)$**-type:** draw a right triangle to convert each angle to $\tan^{-1}$.
5. **Equations:** convert everything to $\tan^{-1}$ (or $\sin^{-1}$) and use the sum formula.

Common mistakes that lose marks

- $\cos^{-1}(-x)=\pi-\cos^{-1}x$, but $\sin^{-1}(-x)=-\sin^{-1}x$ and $\tan^{-1}(-x)=-\tan^{-1}x$; $\cot^{-1}(-x)=\pi-\cot^{-1}x$.
- Writing $\sin^{-1}(\sin\frac{2\pi}{3})=\frac{2\pi}{3}$ (outside the range; answer $\frac{\pi}{3}$).
- Confusing $\sin^{-1}x$ with $\dfrac{1}{\sin x}$.
- Giving answers in degrees when the range is in radians.

Question types: 1-mark principal values and range MCQs; 2–3 marks: simplify using a substitution, evaluate $f^{-1}(f(\theta))$.

2C Exercise-wise solution patterns

Exercise 2.1 (14 questions, 3 patterns)

Pattern A: Principal value of a single inverse function

Solved (Q5): $\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\frac{1}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ (in $[0, \pi]$).
(Q3): $\operatorname{cosec}^{-1} 2 = \frac{\pi}{6}$ since $\operatorname{cosec}\frac{\pi}{6} = 2$.

Pattern B: Combinations of principal values

Solved (Q11): $\tan^{-1} 1 + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6} = \frac{3\pi}{4}$.

Pattern C: Range MCQ

Solved (Q13): $\sin^{-1} x = y \Rightarrow -\frac{\pi}{2} \le y \le \frac{\pi}{2}$ **(B)**.

Pattern A: Q1–10. Pattern B: Q11, 12, 14. Pattern C: Q13.

Exercise 2.2 (15 questions, 4 patterns)

Pattern A: Prove multiple-angle identities

Solved (Q1): put $x = \sin\theta$: $\sin^{-1}(3x - 4x^3) = \sin^{-1}(\sin 3\theta) = 3\theta = 3\sin^{-1} x$ (valid as $3\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ for $x \in [-\frac{1}{2}, \frac{1}{2}]$).

Pattern B: Write in simplest form (substitution or half-angles)

Solved (Q3): $x = \tan\theta$: $\dfrac{\sqrt{1+x^2} - 1}{x} = \dfrac{\sec\theta - 1}{\tan\theta} = \dfrac{1 - \cos\theta}{\sin\theta} = \tan\frac{\theta}{2} \Rightarrow \tan^{-1}(\cdot) = \dfrac{1}{2}\tan^{-1} x$.
(Q4): $\sqrt{\frac{1-\cos x}{1+\cos x}} = \tan\frac{x}{2} \Rightarrow$ answer $\frac{x}{2}$.

Pattern C: $f^{-1}(f(\theta))$ outside the principal range

Solved (Q10): $\sin^{-1}\left(\sin\frac{2\pi}{3}\right) = \sin^{-1}\left(\sin\frac{\pi}{3}\right) = \dfrac{\pi}{3}$.
(Q13): $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \cos^{-1}\left(\cos\frac{5\pi}{6}\right) = \dfrac{5\pi}{6}$ **(B)**.

Pattern D: Evaluate using a right triangle

Solved (Q12): $\sin^{-1}\frac{3}{5} = \tan^{-1}\frac{3}{4}$, $\cot^{-1}\frac{3}{2} = \tan^{-1}\frac{2}{3}$.
$\tan(\cdot) = \dfrac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4}\cdot\frac{2}{3}} = \dfrac{17/12}{1/2} = \dfrac{17}{6}$.

Pattern A: Q1, 2. Pattern B: Q3–7. Pattern C: Q10, 11, 13. Pattern D: Q8, 9, 12, 14, 15.

Miscellaneous Exercise (14 questions, 4 patterns)

Pattern A: $f^{-1}(f(\theta))$

Solved (Q1): $\cos\frac{13\pi}{6} = \cos\frac{\pi}{6} \Rightarrow \cos^{-1}\left(\cos\frac{13\pi}{6}\right) = \frac{\pi}{6}$.

Pattern B: Prove sums of inverse functions

Solved (Q3): let $\theta = \sin^{-1}\frac{3}{5}$, $\tan\theta = \frac{3}{4}$; $\tan 2\theta = \dfrac{2\cdot\frac{3}{4}}{1-\frac{9}{16}} = \dfrac{24}{7}$, so $2\sin^{-1}\frac{3}{5} = \tan^{-1}\frac{24}{7}$.

Pattern C: Prove simplifications (substitution)

Solved (Q8): $x = \tan^2\theta$: $\frac{1}{2}\cos^{-1}\dfrac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1}{2}\cos^{-1}(\cos 2\theta) = \theta = \tan^{-1}\sqrt{x}$.

Pattern D: Equations and MCQs

Solved (Q12): $\tan^{-1}\dfrac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1}x = \frac{1}{2}\tan^{-1}x \Rightarrow \frac{3}{2}\tan^{-1}x = \frac{\pi}{4} \Rightarrow x = \dfrac{1}{\sqrt{3}}$.

Pattern A: Q1, 2. Pattern B: Q3–7. Pattern C: Q8, 9, 10. Pattern D: Q11–14.

2D Pattern spotter

If you see...	Do this	See
Values		
Principal value	Angle inside the principal range	2.1 A
Negative argument	$\sin^{-1}, \tan^{-1}$: odd; $\cos^{-1}, \cot^{-1}$: $\pi-$	2.1 A
$\sin^{-1}(\sin\theta)$ etc.	Move θ into the range first	2.2 C
$\tan(\sin^{-1}a \pm \cot^{-1}b)$	Triangle $\to \tan^{-1}$, then sum formula	2.2 D
Simplifying and proving		
$\sqrt{1+x^2}$, $1+x^2$	$x = \tan\theta$	2.2 B
$\sqrt{1-x^2}$, $3x-4x^3$	$x = \sin\theta$	2.2 A
$\frac{1-x}{1+x}$, $\sqrt{1\pm x}$	$x = \cos\theta$ or $\cos 2\theta$; half-angles	Misc C
Sum of two inverse values	Convert to $\tan^{-1}$, add	Misc B
Equation in $\tan^{-1}$	Use $\tan^{-1}\frac{1-x}{1+x} = \frac{\pi}{4} - \tan^{-1}x$	Misc D

Chapter 3 Matrices

3A JEE basics (extra)

J1. Special matrices

- Idempotent: $A^2 = A$ (then $(I+A)^n = I + (2^n - 1)A$; Misc Q11: $(I+A)^3 - 7A = I$).
- Involutory: $A^2 = I$; nilpotent: $A^k = O$; orthogonal: $AA' = I$.
- Rotation matrix $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$: $R(\alpha)R(\beta) = R(\alpha+\beta)$, $R(\theta)^n = R(n\theta)$.

J2. Trace and counting

$\mathrm{tr}(A+B) = \mathrm{tr}A + \mathrm{tr}B$, $\mathrm{tr}(AB) = \mathrm{tr}(BA)$. Number of $m \times n$ matrices with entries from k values: k^{mn}.

Example (Ex 3.1 Q10): 3×3 with entries 0 or 1: $2^9 = 512$.

J3. Symmetric/skew facts

Diagonal of a skew-symmetric matrix is zero; AB (both symmetric) is symmetric $\iff AB = BA$; $AB - BA$ is skew-symmetric; $B'AB$ keeps the symmetry type of A.

3B Exam tips

Which method?

1. **Order / number of elements:** factor pairs of the number of elements.
2. **Equality of matrices:** same order, equate corresponding entries, solve.
3. **Products:** check order ($m \times n$ times $n \times p$); row $\times$ column.
4. **Matrix equations ($X+Y$, $X-Y$):** add/subtract the equations like ordinary algebra.
5. **Polynomial in A ($A^2 - 5A + 7I$):** compute A^2, then combine.
6. **Symmetric + skew:** $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$.

Common mistakes that lose marks

- Assuming $AB = BA$; matrix multiplication is not commutative.
- $(AB)' = B'A'$, not $A'B'$.
- Writing 5 instead of $5I$ in $A^2 - 5A + 7I$.
- Arithmetic slips in 3×3 products: check one entry again.

Question types: 1-mark: order, type of matrix, MCQs on orders of products; 2–3 marks: products, matrix equations, symmetric/skew split; word problems using matrix multiplication.

3C Exercise-wise solution patterns

Exercise 3.1 (10 questions, 3 patterns)

Pattern A: Order, elements, counting

Solved (Q2): 24 elements: $1\times 24,\ 24\times 1,\ 2\times 12,\ 12\times 2,\ 3\times 8,\ 8\times 3,\ 4\times 6,\ 6\times 4$. 13 elements (prime): $1\times 13,\ 13\times 1$.

Pattern B: Construct a matrix from a rule a_{ij}

Solved (Q4(i)): $a_{ij}=\dfrac{(i+j)^2}{2}$: $\begin{bmatrix} 2 & \frac{9}{2} \\ \frac{9}{2} & 8 \end{bmatrix}$.

Pattern C: Equality of matrices

Solved (Q7): $a-b=-1,\ 2a-b=0\Rightarrow a=1,\ b=2;\ \ 2a+c=5\Rightarrow c=3;\ \ 3c+d=13\Rightarrow d=4.$

Pattern A: Q1, 2, 3, 8, 10. Pattern B: Q4, 5. Pattern C: Q6, 7, 9.

Exercise 3.2 (22 questions, 6 patterns)

Pattern A: Operations and products

Solved (Q1(iv)): $AB=\begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}=\begin{bmatrix} 2-8 & 6+20 \\ 3-4 & 9+10 \end{bmatrix}=\begin{bmatrix} -6 & 26 \\ -1 & 19 \end{bmatrix}$.

Pattern B: Matrix equations for X, Y or unknown entries

Solved (Q7(i)): add: $2X=\begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}\Rightarrow X=\begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$; subtract: $Y=\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$.

Pattern C: Polynomial in a matrix

Solved (Q17): $A^2=\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$, $kA-2I=\begin{bmatrix} 3k-2 & -2k \\ 4k & -2k-2 \end{bmatrix}$. Equating: $-2k=-2\Rightarrow k=1$.

Pattern D: Trigonometric matrices

Solved (Q13): $F(x)F(y)$: the $(1,1)$ entry is $\cos x\cos y-\sin x\sin y=\cos(x+y)$, and similarly every entry, giving $F(x+y)$.

Pattern E: Word problems

Solved (Q19): $x+y=30000,\ 0.05x+0.07y=1800\Rightarrow x=y=$ Rs 15000. (b) for Rs 2000: $x=5000,\ y=25000$.

Pattern F: Orders MCQ

Solved (Q21): PY needs $k = 3$ (order $p \times k$); WY is $n \times k$; to add, $p = n$ **(A)**.

Pattern A: Q1–6, 14. Pattern B: Q7–12. Pattern C: Q15, 16, 17. Pattern D: Q13, 18. Pattern E: Q19, 20. Pattern F: Q21, 22.

Exercise 3.3 (12 questions, 3 patterns)

Pattern A: Transpose properties

Method: compute both sides and compare. **(Q5):** for a column A and row B, check $(AB)' = B'A'$ entry by entry.

Pattern B: Symmetric / skew-symmetric checks

Solved (Q12): $A + A' = \begin{bmatrix} 2\cos\alpha & 0 \\ 0 & 2\cos\alpha \end{bmatrix} = I \Rightarrow \cos\alpha = \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{3}$ **(B)**.

Pattern C: Sum of symmetric and skew-symmetric

Solved (Q10(i)): $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$: $P = \frac{1}{2}(A + A') = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$, $Q = \frac{1}{2}(A - A') = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$, $A = P + Q$.

Pattern A: Q1–6. Pattern B: Q7, 8, 11, 12. Pattern C: Q9, 10.

Exercise 3.4 (1 question)

Q1: A, B are inverses $\iff AB = BA = I$ **(D)**.

Miscellaneous Exercise (11 questions, 4 patterns)

Pattern A: Proofs with transposes

Solved (Q1): $(AB - BA)' = B'A' - A'B' = BA - AB = -(AB - BA)$: skew-symmetric.

Pattern B: Find unknowns from a product equation

Solved (Q4): $[1\ 2\ 1]\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} = [6\ 2\ 4]$; $[6\ 2\ 4]\begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 4 + 4x = 0 \Rightarrow x = -1$.

Pattern C: Polynomial identities and MCQs

Solved (Q9): $A^2 = \begin{bmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{bmatrix} = I \Rightarrow 1 - \alpha^2 - \beta\gamma = 0$ **(C)**.

Pattern D: Word problem with matrices

(Q7): revenue = (sales matrix) × (price column); profit = revenue − cost.

Pattern A: Q1, 2, 10. Pattern B: Q3, 4, 6, 8. Pattern C: Q5, 9, 11. Pattern D: Q7.

3D Pattern spotter

If you see...	Do this	See
Basics		
Possible orders	Factor pairs of the number of elements	3.1 A
a_{ij} = formula	Substitute each i, j	3.1 B
Two equal matrices	Equate entries	3.1 C
Operations		
Product	Row × column; check orders	3.2 A
$X+Y$, $X-Y$ given	Add and subtract the equations	3.2 B
$A^2+kA+cI=O$	Compute A^2, equate entries	3.2 C
Order MCQ	$m\times n$ times $n\times p$	3.2 F
Transpose		
$A=P+Q$	$P=\frac{1}{2}(A+A')$, $Q=\frac{1}{2}(A-A')$	3.3 C
Prove skew/symmetric	Take transpose of the expression	Misc A

Chapter 4 Determinants

4A JEE basics (extra)

J1. Properties of determinants (beyond the current NCERT book)

- $|A'| = |A|$; interchanging two rows changes the sign; two equal rows $\Rightarrow 0$.
- $R_i \to R_i + kR_j$ leaves the value unchanged (use to create zeros before expanding).
- Common factor of a row can be taken out; $|kA| = k^n|A|$ for order n.

Example (Misc Q5): $\begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$: $C_1 \to C_1 + C_2 + C_3$ gives a factor $2(x+y)$; the result is $-2(x^3+y^3)$.

J2. Adjoint results

$|\text{adj}\, A| = |A|^{n-1}$, $\text{adj}(\text{adj}\, A) = |A|^{n-2}A$, $\text{adj}(AB) = \text{adj}B \cdot \text{adj}A$, $|A^{-1}| = \dfrac{1}{|A|}$.

J3. Cramer's rule

$x = \dfrac{\Delta_1}{\Delta}$, $y = \dfrac{\Delta_2}{\Delta}$, $z = \dfrac{\Delta_3}{\Delta}$. If $\Delta = 0$ and all $\Delta_i = 0$: infinitely many solutions (or none); $\Delta = 0$, some $\Delta_i \neq 0$: no solution.

4B Exam tips

Which method?

1. **Evaluate:** expand along the row/column with the most zeros.
2. **Area / collinearity / line:** $\frac{1}{2}|\det|$; collinear $\iff$ det $= 0$; line through two points: put (x, y) as the third point.
3. **Inverse:** $A^{-1} = \dfrac{1}{|A|}\text{adj}\, A$; check $|A| \neq 0$ first.
4. $A^2 + aA + bI = O$ **given:** multiply by A^{-1} to get A^{-1} without cofactors.
5. **System of equations:** write $AX = B$; if $|A| \neq 0$, $X = A^{-1}B$ (unique). If $|A| = 0$: $(\text{adj}A)B \neq O \Rightarrow$ inconsistent; $= O \Rightarrow$ infinitely many or none.

Common mistakes that lose marks

- Sign pattern of cofactors: $(-1)^{i+j}$: $+ - + \,/\, - + - \,/\, + - +$.
- adj A is the *transpose* of the cofactor matrix.
- Area must be positive; if area is given, solve with both $+$ and $-$.
- $|2A| = 2^n|A|$, not $2|A|$.
- In $X = A^{-1}B$, write A^{-1} on the *left*.

Question types: 1-mark: $|kA|$, $|\text{adj}A|$, area MCQ; 3 marks: inverse, area, consistency; 5 marks: solve a system by matrix method (often a word problem), or use the product of two matrices to find an inverse.

4C Exercise-wise solution patterns

Exercise 4.1 (8 questions, 3 patterns)

Pattern A: Evaluate

Method: 2×2: $ad-bc$; 3×3: expand along a row with zeros. **(Q1):** $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = -2+20 = 18.$

Pattern B: $|kA| = k^n|A|$

Solved (Q4): 3×3, so $|3A| = 3^3|A| = 27|A|$ (check: $|A| = 4$, $|3A| = 108$).

Pattern C: Solve for x

Solved (Q7(i)): $2-20 = 2x^2-24 \Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$. **(Q8):** $x^2-36=0 \Rightarrow x = \pm 6$ **(B)**.

Pattern A: Q1, 2, 5, 6. Pattern B: Q3, 4. Pattern C: Q7, 8.

Exercise 4.2 (5 questions, 3 patterns)

Pattern A: Area of a triangle (and finding k)

Solved (Q3(i)): $\frac{1}{2}|k(0-2)+4(2-0)+0| = 4 \Rightarrow |8-2k| = 8 \Rightarrow k = 0$ or 8.
(Q5): $\frac{1}{2}|50-10k| = 35 \Rightarrow k = 12$ or -2 **(D)**.

Pattern B: Collinear points

Solved (Q2): $C_2 \to C_2 + C_1$ makes column 2 all $a+b+c$, proportional to column 3 (all 1): det $= 0$, collinear.

Pattern C: Equation of a line

Solved (Q4(i)): $\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0 \Rightarrow -4x+2y = 0 \Rightarrow y = 2x.$

Pattern A: Q1, 3, 5. Pattern B: Q2. Pattern C: Q4.

Exercise 4.3 (5 questions, 2 patterns)

Pattern A: Minors and cofactors

M_{ij}: delete row i, column j; $A_{ij} = (-1)^{i+j}M_{ij}$. **(Q1(ii)):** $M_{11} = d$, $M_{12} = b$, $M_{21} = c$, $M_{22} = a$; cofactors $d, -b, -c, a$.

Pattern B: Expand using cofactors

Solved (Q3): second row $(2,0,1)$, cofactors $A_{21} = 7$, $A_{22} = 7$, $A_{23} = -7$: $\Delta = 2(7)+0(7)+1(-7) = 7$.

Pattern A: Q1, 2. Pattern B: Q3, 4, 5 (Q5: (D), a row/column times its own cofactors).

Exercise 4.4 (18 questions, 5 patterns)

Pattern A: Adjoint and $A(\text{adj}A) = |A|I$

2×2 shortcut: swap a_{11}, a_{22}; change signs of a_{12}, a_{21}. **(Q1):** $\text{adj}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$.

Pattern B: Inverse by adjoint

Solved (Q7): $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$, $|A| = 10$, $\text{adj}A = \begin{bmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{bmatrix}$, $A^{-1} = \frac{1}{10}\text{adj}A$.

Pattern C: Verify $(AB)^{-1} = B^{-1}A^{-1}$

Method (Q12): find AB and its inverse; find B^{-1}, A^{-1}, multiply in that order; compare.

Pattern D: Inverse from a matrix polynomial

Solved (Q13): $A^2 - 5A + 7I = O$. Multiply by A^{-1}: $A - 5I + 7A^{-1} = O \Rightarrow A^{-1} = \frac{1}{7}(5I - A) = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$.

(Q14): $A^2 = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$; $11 + 3a + b = 0$, $8 + 2a = 0 \Rightarrow a = -4$, $b = 1$.

Pattern E: Properties MCQ

Solved (Q17, 18): $|\text{adj}A| = |A|^{n-1} = |A|^2$ **(B)**; $\det(A^{-1}) = \frac{1}{\det A}$ **(B)**.

Pattern A: Q1–4. Pattern B: Q5–11. Pattern C: Q12. Pattern D: Q13–16. Pattern E: Q17, 18.

Exercise 4.5 (16 questions, 4 patterns)

Pattern A: Consistency

Solved (Q3): $\begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 0$; $(\text{adj}A)B = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \neq O$: **inconsistent**.

Pattern B: Solve by matrix method

Solved (Q8): $2x - y = -2$, $3x + 4y = 3$: $|A| = 11$, $X = \frac{1}{11}\begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -5/11 \\ 12/11 \end{bmatrix}$.

Pattern C: Find A^{-1}, then solve

(Q15): the coefficient matrix is exactly A; $X = A^{-1}B$ gives $x = 1$, $y = 2$, $z = 3$.

Pattern D: Word problem

Solved (Q16): $4x + 3y + 2z = 60$, $2x + 4y + 6z = 90$, $6x + 2y + 3z = 70$; $|A| = 50$: onion Rs 5, wheat Rs 8, rice Rs 8 per kg.

Pattern A: Q1–6. Pattern B: Q7–14. Pattern C: Q15. Pattern D: Q16.

Miscellaneous Exercise (9 questions, 4 patterns)

Pattern A: Evaluate / independent of a variable

Solved (Q1): expanding gives $-x^3$ (all θ terms cancel by $\sin^2\theta + \cos^2\theta = 1$): independent of θ.

Pattern B: Inverse properties

(Q3): $(AB)^{-1} = B^{-1}A^{-1}$: find B^{-1} and multiply by the given A^{-1}.

Pattern C: Reciprocal substitution

Solved (Q7): $u = \frac{1}{x}$, $v = \frac{1}{y}$, $w = \frac{1}{z}$: $2u + 3v + 10w = 4$, $4u - 6v + 5w = 1$, $6u + 9v - 20w = 2$; $|A| = 1200$; $u = \frac{1}{2}$, $v = \frac{1}{3}$, $w = \frac{1}{5} \Rightarrow x = 2$, $y = 3$, $z = 5$.

Pattern D: MCQs

Solved (Q9): $\det A = 2 + 2\sin^2\theta \in [2, 4]$ **(D)**. **(Q8):** inverse of a diagonal matrix: reciprocals on the diagonal **(A)**.

Pattern A: Q1, 2, 5, 6. Pattern B: Q3, 4. Pattern C: Q7. Pattern D: Q8, 9.

4D Pattern spotter

If you see...	Do this	See
Values		
Evaluate a determinant	Expand along the row with most zeros	4.1 A
$\lvert kA\rvert$	$k^n\lvert A\rvert$	4.1 B
Equation with determinants	Expand both sides, solve	4.1 C
Geometry		
Area of triangle	$\frac{1}{2}\lvert\det\rvert$; $\pm$ if area given	4.2 A
Collinear points	$\det = 0$	4.2 B
Line through two points	(x, y) as third point, $\det = 0$	4.2 C

If you see…	Do this	See
Inverse and systems		
Cofactors	$(-1)^{i+j}M_{ij}$	4.3 A
Inverse	$\frac{1}{\lvert A\rvert}\text{adj}A$	4.4 B
$A^2 + aA + bI = O$	Multiply by A^{-1}	4.4 D
Consistent or not	$\lvert A\rvert$, then $(\text{adj}A)B$	4.5 A
Solve a system	$X = A^{-1}B$	4.5 B
$\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ in equations	Substitute u, v, w	Misc C

Chapter 5 Continuity and Differentiability

5A JEE basics (extra)

J1. Standard limits used in continuity

$\lim_{x\to0}\frac{\sin x}{x}=1,\ \lim_{x\to0}\frac{1-\cos x}{x^2}=\frac{1}{2},\ \lim_{x\to0}\frac{e^x-1}{x}=1,\ \lim_{x\to0}\frac{\log(1+x)}{x}=1,\ \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}.$

Example (Ex 5.1 Q26): put $x=\frac{\pi}{2}+h$: $\frac{k\cos x}{\pi-2x}=\frac{-k\sin h}{-2h}\to\frac{k}{2}=3\Rightarrow k=6.$

J2. Rolle's theorem and Mean Value Theorem

Rolle: f continuous on $[a,b]$, differentiable on (a,b), $f(a)=f(b)\Rightarrow f'(c)=0$ for some $c\in(a,b)$.

MVT: $f'(c)=\dfrac{f(b)-f(a)}{b-a}$. **Example:** $f=x^2$ on $[2,4]$: $2c=\frac{16-4}{2}=6\Rightarrow c=3$.

J3. Differentiability shortcuts

$|x-a|$ is not differentiable at a; $[x]$ is discontinuous at integers; $x^n|x|$ is differentiable n times at 0 ($|x|^3$ has f'' everywhere: Misc Q18, $f''=6|x|$).

$\dfrac{d}{dx}|f(x)|=\dfrac{f(x)}{|f(x)|}f'(x)$ where $f(x)\neq0$.

5B Exam tips

Which method?

1. **Continuity at $x=a$:** LHL = RHL = $f(a)$. For piecewise functions check only the joining points; each piece (polynomial, sin, ...) is continuous elsewhere.
2. **Find k:** equate the two one-sided limits (or limit = $f(a)$).
3. **Differentiability at a:** LHD = RHD. Differentiable $\Rightarrow$ continuous, not conversely.
4. **Composite:** chain rule from outside in.
5. **Implicit:** differentiate both sides, collect $\frac{dy}{dx}$.
6. **Inverse trig:** substitute ($x=\tan\theta$, $\sin\theta$, $\cos\theta$) to simplify first.
7. **Variable power u^v, or long products/quotients:** take log first.
8. **Parametric:** $\frac{dy}{dx}=\frac{dy/dt}{dx/dt}$.
9. **Prove a second-order relation:** find y_1, clear denominators, differentiate again.

Common mistakes that lose marks

- Not checking $f(a)$ itself (a function can have equal limits but wrong value).
- $\frac{d}{dx}(x^x)\neq x\cdot x^{x-1}$; use logarithms: $x^x(1+\log x)$.
- For u^v+w^z, differentiate each term separately (log of a sum is not allowed).
- Second parametric derivative: $\dfrac{d^2y}{dx^2}=\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)\cdot\dfrac{dt}{dx}$, not $\frac{y''(t)}{x''(t)}$.
- Forgetting the condition (e.g. $|x|<1$) when simplifying with $x=\tan\theta$.

Question types: 1-mark: continuity of standard functions, simple chain rule; 2–3 marks: find k, implicit/parametric derivatives, y^x-type; 5 marks: logarithmic differentiation, prove a second-order relation.

5C Exercise-wise solution patterns

Exercise 5.1 (34 questions, 4 patterns)

Pattern A: Continuity of standard / combined functions

Solved (Q2): $\lim_{x\to 3}(2x^2-1) = 17 = f(3)$: continuous at $x = 3$. (Polynomials, sin, cos, $|x|$ and their sums, products, compositions are continuous.)

Pattern B: Piecewise functions (check joining points)

Solved (Q6): $f = 2x+3$ $(x \le 2)$, $2x-3$ $(x > 2)$. At $x = 2$: LHL $= 7$, RHL $= 1$: discontinuous. Elsewhere continuous. Only discontinuity: $x = 2$.

Pattern C: Find constants for continuity

Solved (Q30): at $x = 2$: $5 = 2a + b$; at $x = 10$: $10a + b = 21$. $a = 2$, $b = 1$.
(Q17): $3a + 1 = 3b + 3 \Rightarrow a = b + \frac{2}{3}$.

Pattern D: Special functions ($[x]$, $x^2 \sin\frac{1}{x}$, $|x|$ combinations)

Solved (Q24): $|x^2 \sin\frac{1}{x}| \le x^2 \to 0$, so the limit at 0 is $0 = f(0)$: continuous everywhere.

Pattern A: Q1–4, 20, 21, 22, 31, 32, 33. Pattern B: Q5–16, 23, 25, 34. Pattern C: Q17, 18, 26–30. Pattern D: Q19, 24.

Exercise 5.2 (10 questions, 2 patterns)

Pattern A: Chain rule

Solved (Q4): $\dfrac{d}{dx}\sec(\tan\sqrt{x}) = \sec(\tan\sqrt{x})\tan(\tan\sqrt{x}) \cdot \sec^2\sqrt{x} \cdot \dfrac{1}{2\sqrt{x}}$.

Pattern B: Not differentiable at a point

Solved (Q9): $f = |x-1|$ at 1: LHD $= \lim_{h\to 0^-}\dfrac{|h|}{h} = -1$, RHD $= +1$: not differentiable.

Pattern A: Q1–8. Pattern B: Q9, 10.

Exercise 5.3 (15 questions, 2 patterns)

Pattern A: Implicit differentiation

Solved (Q5): $x^2 + xy + y^2 = 100 \Rightarrow 2x + y + xy' + 2yy' = 0 \Rightarrow y' = -\dfrac{2x+y}{x+2y}$.

Pattern B: Inverse trig with a substitution

Solved (Q9): $x = \tan\theta$: $\sin^{-1}\dfrac{2x}{1+x^2} = \sin^{-1}(\sin 2\theta) = 2\theta = 2\tan^{-1}x \Rightarrow y' = \dfrac{2}{1+x^2}$ (for $|x| < 1$).
(Q15): $\sec^{-1}\dfrac{1}{2x^2-1} = \cos^{-1}(2x^2-1)$; $x = \cos\theta$ gives $2\cos^{-1}x$, so $y' = -\dfrac{2}{\sqrt{1-x^2}}$.

Pattern A: Q1–8. Pattern B: Q9–15.

Exercise 5.4 (10 questions, 1 pattern)

Pattern A: Exponential and logarithmic functions (chain/quotient rule)

Solved (Q1): $\dfrac{d}{dx}\dfrac{e^x}{\sin x} = \dfrac{e^x \sin x - e^x \cos x}{\sin^2 x} = \dfrac{e^x(\sin x - \cos x)}{\sin^2 x}.$

(Q8): $\dfrac{d}{dx}\log(\log x) = \dfrac{1}{\log x}\cdot\dfrac{1}{x} = \dfrac{1}{x\log x}.$

Pattern A: Q1–10.

Exercise 5.5 (18 questions, 3 patterns)

Pattern A: Long products / quotients (take log)

Solved (Q16): $f = (1+x)(1+x^2)(1+x^4)(1+x^8)$; $\dfrac{f'}{f} = \dfrac{1}{1+x} + \dfrac{2x}{1+x^2} + \dfrac{4x^3}{1+x^4} + \dfrac{8x^7}{1+x^8}.$

At $x=1$: $f = 16$, $\frac{f'}{f} = \frac{1}{2} + 1 + 2 + 4 = \frac{15}{2} \Rightarrow f'(1) = 120.$

Pattern B: Variable base and power, u^v (and sums of them)

Solved (Q3): $y = (\log x)^{\cos x}$: $\log y = \cos x \log(\log x)$

$\dfrac{y'}{y} = -\sin x \log(\log x) + \dfrac{\cos x}{x\log x} \Rightarrow y' = (\log x)^{\cos x}\left[\dfrac{\cos x}{x\log x} - \sin x \log(\log x)\right].$

Pattern C: Implicit equations with powers

Solved (Q15): $xy = e^{x-y} \Rightarrow \log x + \log y = x - y \Rightarrow \dfrac{1}{x} + \dfrac{y'}{y} = 1 - y' \Rightarrow y' = \dfrac{y(x-1)}{x(y+1)}.$

Pattern A: Q1, 2, 5, 16, 17, 18. Pattern B: Q3, 4, 6–11. Pattern C: Q12–15.

Exercise 5.6 (11 questions, 1 pattern)

Pattern A: Parametric differentiation

Solved (Q6): $x = a(\theta - \sin\theta)$, $y = a(1+\cos\theta)$: $\dfrac{dy}{dx} = \dfrac{-a\sin\theta}{a(1-\cos\theta)} = \dfrac{-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = -\cot\dfrac{\theta}{2}.$

Pattern A: Q1–11.

Exercise 5.7 (17 questions, 2 patterns)

Pattern A: Find the second derivative

Solved (Q5): $y = x^3\log x$: $y' = 3x^2\log x + x^2$, $y'' = 6x\log x + 3x + 2x = x(6\log x + 5)$.

Pattern B: Prove a relation between y, y_1, y_2

Solved (Q13): $y = 3\cos(\log x) + 4\sin(\log x)$: $xy_1 = -3\sin(\log x) + 4\cos(\log x)$.

Differentiate: $xy_2 + y_1 = \dfrac{-3\cos(\log x) - 4\sin(\log x)}{x} = -\dfrac{y}{x} \Rightarrow x^2y_2 + xy_1 + y = 0.$

Pattern A: Q1–10, 12. Pattern B: Q11, 13–17.

Miscellaneous Exercise (22 questions, 4 patterns)

Pattern A: Mixed derivatives

Solved (Q8): $\dfrac{d}{dx}\cos(a\cos x + b\sin x) = (a\sin x - b\cos x)\sin(a\cos x + b\sin x)$.

Pattern B: Parametric / implicit

Solved (Q12): $\dfrac{dy}{dx} = \dfrac{12\sin t}{10(1-\cos t)} = \dfrac{6}{5}\cot\dfrac{t}{2}$.

Pattern C: Second derivative proofs

Method (Q22): $y = e^{a\cos^{-1}x}$: $\sqrt{1-x^2}\,y_1 = -ay$; square, differentiate, divide by $2y_1$.

Pattern D: Conceptual

Solved (Q20): yes: $f(x) = |x-1| + |x-2|$ is continuous everywhere but not differentiable at $x = 1, 2$.

Pattern A: Q1–11. Pattern B: Q12, 13, 14, 16. Pattern C: Q15, 17, 22. Pattern D: Q18–21.

5D Pattern spotter

If you see...	Do this	See
Continuity		
Standard / combined function	Continuous; check lim $= f(a)$ if asked	5.1 A
Piecewise	LHL, RHL, $f(a)$ at joining points	5.1 B
Find k (or a, b)	Equate one-sided limits	5.1 C
$[x]$, $x^2\sin\frac{1}{x}$	Integers / squeeze argument	5.1 D
Differentiability and rules		
Is it differentiable at a?	LHD = RHD	5.2 B
Function of a function	Chain rule	5.2 A
x and y mixed	Implicit differentiation	5.3 A
$\sin^{-1}\frac{2x}{1+x^2}$ etc.	Substitute, simplify, then differentiate	5.3 B
Special methods		
Long product / quotient	Log differentiation	5.5 A
u^v	$\log y = v\log u$	5.5 B
$x = f(t)$, $y = g(t)$	$\frac{dy/dt}{dx/dt}$	5.6 A
Prove $x^2y_2 + xy_1 + y = 0$-type	Clear denominators before differentiating again	5.7 B

Chapter 6 Application of Derivatives

6A JEE basics (extra)

J1. Tangents and normals (not in the current board syllabus; needed for JEE)

Slope of tangent at (x_1, y_1): $m = \left.\dfrac{dy}{dx}\right|_{(x_1,y_1)}$. Tangent: $y - y_1 = m(x - x_1)$; normal: $y - y_1 = -\dfrac{1}{m}(x - x_1)$.

Example: $y = x^2$ at $(1, 1)$: $m = 2$. Tangent $y = 2x - 1$; normal $x + 2y = 3$.

J2. Approximations

$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$. **Example:** $\sqrt{25.3} \approx 5 + \dfrac{0.3}{2\sqrt{25}} = 5.03$.

J3. Standard optimisation results (useful for MCQs)

Rectangle in a circle, max area	a square
Open box from a square sheet of side a	cut squares of side $\frac{a}{6}$
Cylinder of max volume in a sphere of radius R	height $\frac{2R}{\sqrt{3}}$
Cone of max volume in a sphere of radius R	height $\frac{4R}{3}$, volume $\frac{8}{27}$ of the sphere
Closed cylinder, fixed volume, least surface	$h = 2r$ (height = diameter)
Cone of fixed slant height, max volume	semi-vertical angle $\tan^{-1}\sqrt{2}$

J4. AM–GM shortcut for $x^m y^n$ with $x + y$ fixed

Maximum of $x^m y^n$ when $x + y = k$ is at $\dfrac{x}{m} = \dfrac{y}{n}$, i.e. $x = \dfrac{mk}{m+n}$, $y = \dfrac{nk}{m+n}$.

Example: $x + y = 60$, maximise xy^3: $\frac{x}{1} = \frac{y}{3} \Rightarrow x = 15$, $y = 45$.

6B Exam tips

Which method should I use? (check in this order)

1. **"How fast", "rate", "per second"** → related rates: write the formula, differentiate with respect to t, then substitute.
2. **"Marginal"** → just differentiate $C(x)$ or $R(x)$ and substitute x.
3. **"Show increasing/decreasing"** → prove the sign of $f'(x)$ on the given interval.
4. **"Find the intervals"** → roots of $f'(x) = 0$, sign table.
5. **"Local max/min"** → critical points, then second derivative test (first derivative test if $f'' = 0$).
6. **"Absolute" or a closed interval** $[a, b]$ → compare f at critical points *and* end points.
7. **Word problem** → one-variable function, $f' = 0$, confirm with f''.
8. **| |, squares, sin/cos without calculus** → use bounds (formula sheet) directly.

Common mistakes that lose marks

- Substituting the given value *before* differentiating in related rates.
- Forgetting the minus sign for a decreasing quantity (e.g. "decreasing at 5 cm/min" means $\frac{dx}{dt} = -5$).
- Forgetting units (cm^2/s, cm^3/s) in rate answers.
- Missing the end points when finding absolute max/min.
- Not checking f'' (or the sign change) to justify max or min.
- Writing "increasing in $x > 2$" instead of an interval $(2, \infty)$.
- In word problems, giving x when the question asks for the area / volume / dimensions.
- Using degrees in calculus with trig: always radians.

Smart exam habits

- Draw a neat labelled figure for every word problem and related-rate question.
- Show the sign table for intervals; examiners give marks for it.
- State clearly: "$f''(c) < 0$, hence $x = c$ is a point of local maximum."
- Check your answer makes sense (a length must be positive, a box side less than half the sheet).

Question types to practise

- **Short / MCQ:** rate of change of area/volume, marginal revenue, interval MCQs, max of $a \sin x + b \cos x$.
- **Medium:** related rates (ladder, cone, balloon), intervals of increase/decrease, local extrema.
- **Long (5 marks / case study):** optimisation word problems (box, cylinder, cone in sphere, window, tank), absolute extrema on an interval.
- **Must-do from NCERT:** Exercise 6.3 Q17–26 and Miscellaneous Q5–8, 12, 14, 15.

6C Exercise-wise solution patterns

Exercise 6.1 (18 questions, 5 patterns)

Pattern A: Rate of change with respect to the variable itself

Spot: "rate of change of A with respect to r" (no time involved). **Method:** just differentiate and substitute.

Solved (Q1): $A = \pi r^2 \Rightarrow \dfrac{dA}{dr} = 2\pi r$. (a) $r = 3$: 6π cm^2/cm; (b) $r = 4$: 8π cm^2/cm.

Pattern B: Related rates with a single formula

Spot: one rate given, another asked, connected by an area/volume formula.

Solved (Q2): cube: $V = x^3$, $\dfrac{dV}{dt} = 3x^2\dfrac{dx}{dt} = 8 \Rightarrow \dfrac{dx}{dt} = \dfrac{8}{3(144)} = \dfrac{1}{54}$ cm/s at $x = 12$.

Surface $S = 6x^2 \Rightarrow \dfrac{dS}{dt} = 12x\dfrac{dx}{dt} = 12(12)\cdot\dfrac{1}{54} = \dfrac{8}{3}$ cm^2/s.

Pattern C: Related rates needing an extra relation (Pythagoras, similar shapes)

Spot: ladder, cone with fixed ratio, shadow. **Method:** use the relation to remove one variable first.

Solved (Q14): $h = \frac{r}{6} \Rightarrow r = 6h$. $V = \frac{1}{3}\pi r^2 h = 12\pi h^3$; $\dfrac{dV}{dt} = 36\pi h^2\dfrac{dh}{dt}$.

$12 = 36\pi(16)\dfrac{dh}{dt} \Rightarrow \dfrac{dh}{dt} = \dfrac{1}{48\pi}$ cm/s.

(Q10, ladder): $x^2 + y^2 = 25$; at $x = 4$, $y = 3$: $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0 \Rightarrow \dfrac{dy}{dt} = -\dfrac{4(2)}{3} = -\dfrac{8}{3}$: height decreases at $\frac{8}{3}$ cm/s.

Pattern D: Points on a curve with a rate condition

Solved (Q11): $6y = x^3 + 2 \Rightarrow 6\dfrac{dy}{dt} = 3x^2\dfrac{dx}{dt}$. With $\frac{dy}{dt} = 8\frac{dx}{dt}$: $48 = 3x^2 \Rightarrow x = \pm 4$.

Points: $(4, 11)$ and $\left(-4, -\frac{31}{3}\right)$.

Pattern E: Marginal cost and marginal revenue

Solved (Q15): MC $= C'(x) = 0.021x^2 - 0.006x + 15$; at $x = 17$: $6.069 - 0.102 + 15 =$ Rs 20.967.
(Q18): MR $= 6x + 36$; at $x = 15$: 126 **(D)**.

Pattern A: Q1, 9, 13, 17. Pattern B: Q2–8, 12. Pattern C: Q10, 14. Pattern D: Q11. Pattern E: Q15, 16, 18.

Exercise 6.2 (19 questions, 5 patterns)

Pattern A: Show increasing / decreasing (sign of f')

Solved (Q18): $f'(x) = 3x^2 - 6x + 3 = 3(x-1)^2 \geq 0$, and $= 0$ only at $x = 1$. Hence f is increasing on **R**.

Pattern B: Find intervals of increase/decrease

Solved (Q5): $f'(x) = 6x^2 - 6x - 36 = 6(x-3)(x+2)$.
Increasing on $(-\infty, -2) \cup (3, \infty)$; decreasing on $(-2, 3)$.

Pattern C: Function involving log or a fraction (simplify f' fully)

Solved (Q7): $y = \log(1+x) - \dfrac{2x}{2+x}$. $y' = \dfrac{1}{1+x} - \dfrac{4}{(2+x)^2} = \dfrac{(2+x)^2 - 4(1+x)}{(1+x)(2+x)^2} = \dfrac{x^2}{(1+x)(2+x)^2}$.
For $x > -1$: numerator ≥ 0 and denominator > 0, so $y' \geq 0$: increasing throughout the domain.

Pattern D: Find a parameter so that f is increasing

Solved (Q14): $f'(x) = 2x + a \geq 0$ on $[1, 2]$. The smallest value of $2x + a$ there is $2 + a$, so $2 + a \geq 0 \Rightarrow a \geq -2$.

Pattern E: MCQs on intervals

Solved (Q19): $y = x^2 e^{-x}$, $y' = x(2-x)e^{-x}$; positive only for $0 < x < 2$: **(D)** $(0, 2)$.

Pattern A: Q1, 2, 3, 10, 11, 15, 16, 17, 18. Pattern B: Q4, 5, 6, 8. Pattern C: Q7, 9. Pattern D: Q14. Pattern E: Q12, 13, 19.

Exercise 6.3 (29 questions, 6 patterns)

Pattern A: Max/min without calculus (squares, modulus, sin/cos bounds)

Solved (Q1(i)): $f(x) = (2x-1)^2 + 3 \geq 3$: minimum 3 at $x = \frac{1}{2}$; no maximum.
(Q2(iv)): $-1 \leq \sin 4x \leq 1 \Rightarrow 2 \leq \sin 4x + 3 \leq 4$: maximum 4, minimum 2.

Pattern B: Local maxima/minima (derivative tests)

Solved (Q3(v)): $f(x) = x^3 - 6x^2 + 9x + 15$, $f' = 3(x-1)(x-3)$, $f'' = 6x - 12$.
$f''(1) = -6 < 0$: local max $f(1) = 19$. $f''(3) = 6 > 0$: local min $f(3) = 15$.
(Q4): if $f'(x)$ is never zero (e.g. e^x, $\frac{1}{x}$, $3x^2 + 2x + 1 > 0$), there is no maximum or minimum.

Pattern C: Absolute max/min on a closed interval

Solved (Q7): $f'(x) = 12x^3 - 24x^2 + 24x - 48 = 12(x-2)(x^2+2)$: only critical point $x = 2$.
$f(0) = 25$, $f(2) = -39$, $f(3) = 16$. Maximum 25 at $x = 0$; minimum -39 at $x = 2$.
(Q11): max at $x = 1 \Rightarrow f'(1) = 4 - 124 + a = 0 \Rightarrow a = 120$.

Pattern D: Numbers with a fixed sum (and profit functions)

Solved (Q15): $x + y = 35$; maximise $f = x^2(35-x)^5$.
$f' = x(35-x)^4[2(35-x) - 5x] = x(35-x)^4(70 - 7x) = 0 \Rightarrow x = 10$. So $x = 10$, $y = 25$.

Pattern E: Geometry optimisation (box, cylinder, cone, wire)

Solved (Q17): cut squares of side x: $V = x(18-2x)^2$; $V' = (18-2x)(18-6x) = 0 \Rightarrow x = 3$ (or 9, which gives no box).
$V''(3) = -72 < 0$: maximum. Cut squares of side 3 cm.

(Q21): $\pi r^2 h = 100$, $S = 2\pi r^2 + \dfrac{200}{r}$; $S' = 4\pi r - \dfrac{200}{r^2} = 0 \Rightarrow r = \left(\dfrac{50}{\pi}\right)^{1/3}$, and then $h = 2r$.

Pattern F: MCQs (nearest point, min/max of a function)

Solved (Q27): point $(x, \frac{x^2}{2})$; $D = x^2 + \left(\frac{x^2}{2} - 5\right)^2$, $D' = x(x^2 - 8) = 0 \Rightarrow x = \pm 2\sqrt{2}$, $y = 4$: **(A)** $(2\sqrt{2}, 4)$.

Pattern A: Q1, 2, 9. Pattern B: Q3, 4. Pattern C: Q5, 7, 8, 10, 11, 12. Pattern D: Q6, 13–16. Pattern E: Q17–26. Pattern F: Q27, 28, 29.

Miscellaneous Exercise (16 questions, 5 patterns)

Pattern A: Intervals of increase/decrease (harder functions)

Solved (Q4): $f'(x) = 3x^2 - \dfrac{3}{x^4} = \dfrac{3(x^6 - 1)}{x^4}$.
Increasing on $(-\infty, -1) \cup (1, \infty)$; decreasing on $(-1, 0) \cup (0, 1)$.
(Q3): $f'(x) = \dfrac{\cos x(4 - \cos x)}{(2 + \cos x)^2}$; since $4 - \cos x > 0$, the sign is that of $\cos x$.

Pattern B: Related rates

Solved (Q16): $V = \pi(10)^2 h \Rightarrow \dfrac{dV}{dt} = 100\pi \dfrac{dh}{dt} = 314 \Rightarrow \dfrac{dh}{dt} = \dfrac{314}{100 \times 3.14} = 1$ m/h **(A)**.

Pattern C: Local max/min and point of inflexion

Solved (Q10): $f'(x) = (x - 2)^3(x + 1)^2(7x - 2)$; critical points -1, $\frac{2}{7}$, 2.
At $x = -1$: no sign change ($(x + 1)^2$): **inflexion**. At $\frac{2}{7}$: $+ \to -$: **local max**. At 2: $- \to +$: **local min**.

Pattern D: Absolute extrema with trig functions

Solved (Q11): $f'(x) = -2 \sin x \cos x + \cos x = \cos x(1 - 2 \sin x) = 0 \Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.
$f(0) = 1$, $f(\frac{\pi}{6}) = \frac{5}{4}$, $f(\frac{\pi}{2}) = 1$, $f(\frac{5\pi}{6}) = \frac{5}{4}$, $f(\pi) = 1$. Max $\frac{5}{4}$ at $x = \frac{\pi}{6}, \frac{5\pi}{6}$; min 1 at $x = 0, \frac{\pi}{2}, \pi$.

Pattern E: Geometry optimisation

Solved (Q8): rectangle width x, height y, semicircle of radius $\frac{x}{2}$. Perimeter $x + 2y + \frac{\pi x}{2} = 10$.
$A = xy + \dfrac{\pi x^2}{8} = 5x - \dfrac{x^2}{2} - \dfrac{\pi x^2}{8}$; $A' = 5 - x - \dfrac{\pi x}{4} = 0 \Rightarrow x = \dfrac{20}{\pi + 4}$, $y = \dfrac{10}{\pi + 4}$ m.
(Q6): $xy \cdot 2 = 8 \Rightarrow xy = 4$; cost $= 280 + 180(x + y)$, least when $x = y = 2$: Rs 1000.

Pattern A: Q3, 4, 13. Pattern B: Q2, 16. Pattern C: Q1, 10. Pattern D: Q11. Pattern E: Q5–9, 12, 14, 15.

6D Pattern spotter

If you see...	Do this	See
1. Rate of change		
"Rate of change of A w.r.t. r"	Differentiate, substitute	6.1 A
"How fast... per second"	Formula $\to \frac{d}{dt}$ both sides $\to$ substitute	6.1 B
Ladder, shadow, cone with $h = kr$	Use the extra relation first, then $\frac{d}{dt}$	6.1 C
"y changes k times as fast as x"	$\frac{dy}{dt} = k\frac{dx}{dt}$ in the differentiated curve	6.1 D
"Marginal cost/revenue"	$C'(x)$ or $R'(x)$ at the given x	6.1 E
Quantity decreasing	Use a negative rate	6B
2. Increasing and decreasing		
"Show increasing on **R**"	Prove $f' \geq 0$ (e.g. a perfect square)	6.2 A
"Find the intervals"	Roots of $f' = 0$, sign table	6.2 B
log or fraction functions	Combine f' into one fraction, check signs	6.2 C
"Find a so that f is increasing"	Smallest value of f' on the interval ≥ 0	6.2 D
Trig functions on $[0, 2\pi]$	Roots of f' in the interval, test points	Misc A
3. Maxima and minima		
Square, $\lvert\ \rvert$, sin, cos	Use bounds; no calculus needed	6.3 A
"Local max/min"	$f' = 0$, then f'' sign	6.3 B
$f''(c) = 0$ or "inflexion"	First derivative test (sign change)	Misc C
f' never zero	No max or min	6.3 B
Closed interval $[a, b]$	Compare critical points and end points	6.3 C
"Max at $x = c$, find a"	$f'(c) = 0$	6.3 C
4. Word problems		
Fixed sum, product/power maximum	One variable, $f' = 0$ (or AM–GM)	6.3 D
Box from a sheet	$V = x(l - 2x)(b - 2x)$	6.3 E
Cylinder/cone, fixed volume or surface	Express in r only; S' or $V' = 0$	6.3 E
Window, tank cost	Perimeter/volume condition gives one variable	Misc E
Nearest point on a curve	Minimise $D =$ distance^2	6.3 F